

Below is our STEM concept list based on our priorities from element C:

Durability

STEM Concept: Science

This applies to our design because it all depends on what materials will not rust, mold, or collect mildew.

Cost effectiveness

STEM Concept: Math

This ties to math because we have to figure out how to make this product as affordable as possible.

All purpose

STEM Concept: Engineering

This applies to our design because we have to figure out a way to make this shower head work in any shower and be easy to use.

Easy to package

STEM Concept: engineering

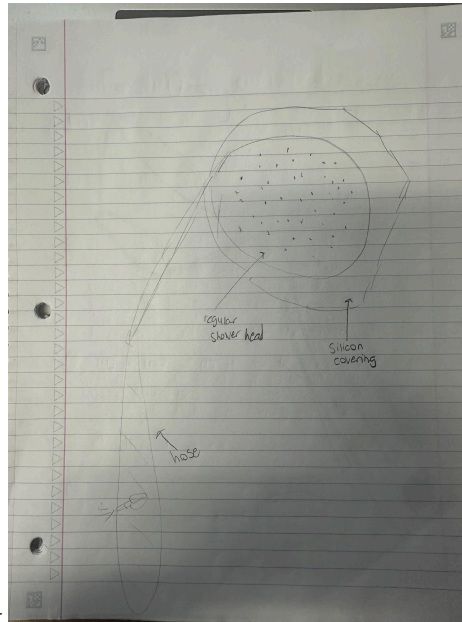
This is tied to our design because it is what makes the manufacturing process easier. If the design is difficult or bulky then it would be hard to package.

Easy to install

STEM Concept: Technology

This affects our design because we want it to be easy and quick to install. It is related to technology because it depends on how many wires or loose ends would need to be attached.

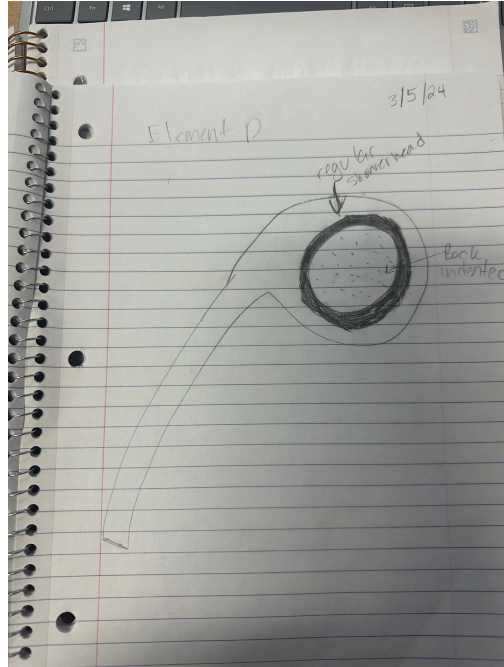
Below are our initial drawings with respect to the STEM concepts we needed to be using:



STEM Principle for Design Solution #1-

STEM Concept: science, math, and engineering

Describe how the STEM concept applies directly to the specifics of your solution: We have to use science with the water pressure and what mineral we can use for the ring around it. We have to use math with the measurements of the ring and to see how it is going to fit around the shower head. Finally we need engineering because we need to figure out what the funnel should be made out of so it will not rust or mold.

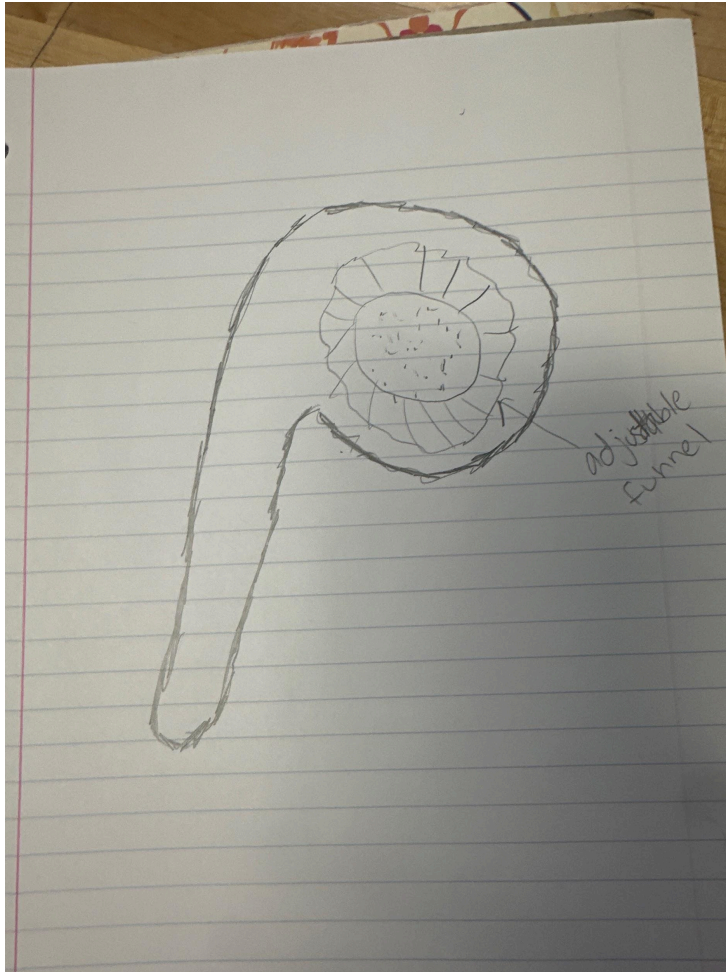


STEM Principle for Design Solution #2-

STEM Concept: we will also be using science and math and engineering

Describe how the STEM concept applies directly to the specifics of your solution: We will be using science because we still need to find out what kind of water pressure we will be having. We also will begin using maining math because we need the dementain of the shower head so we know how far to go in for the indent of the shower head and we also need measurements. Finally we need engineering so we can figure out how to make the indent into the shower head and find a way the water can flow through it without it getting backed up and so the water can go through easily but that could create more water pressure so that could be a good thing.

STEM Principle for Design Solution #3



STEM Concept: math and engineering

Describe how the STEM concept applies directly to the specifics of your solution: we will be using math with the adjustable funnel to see how we can make it adjustable and to see how big or small to make it and also what the water pressure is going to be with the funnel on it. Finally we need engineering so we can figure out how many funnels are adjustable and what material we need to make it out of.